



Discrete Unit Back-translation for Speech Translation

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End-to-End Speech Translation (ST)



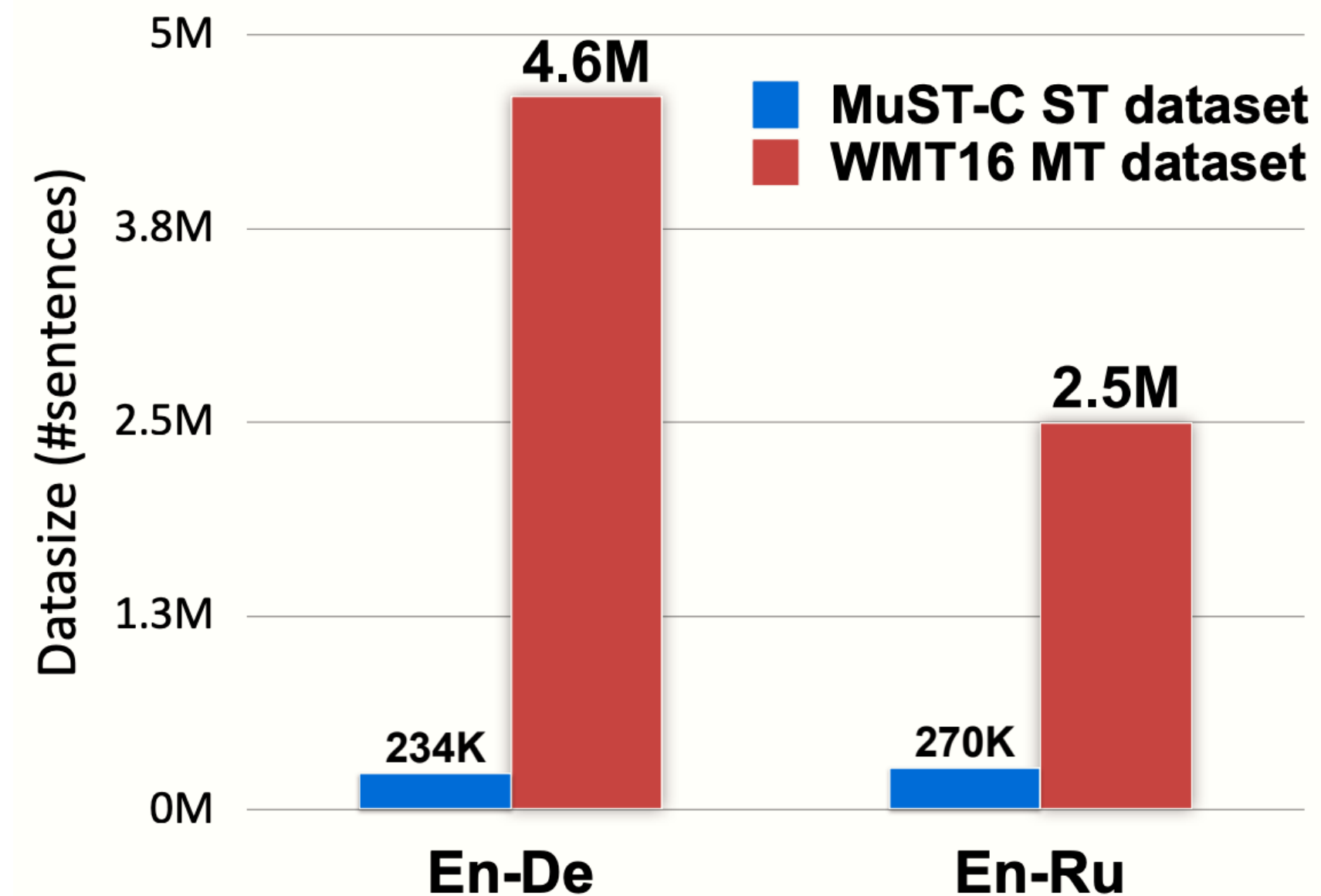
Unwritten / dialect
communication



Building E2E-ST model is challenging

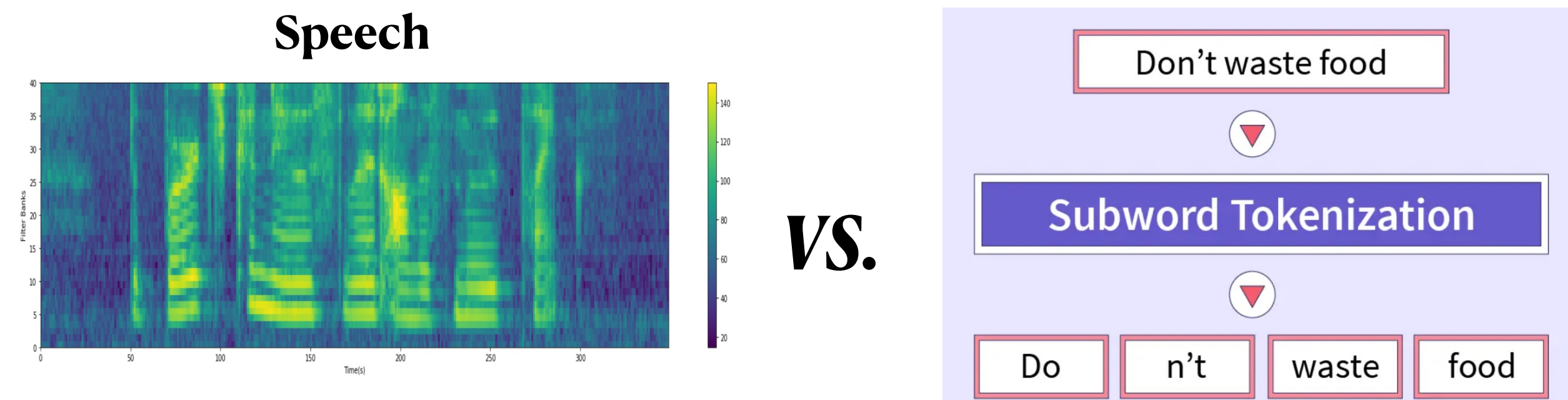
- **Challenge1: Data Scarcity**

- Similar to **low-resource MT**



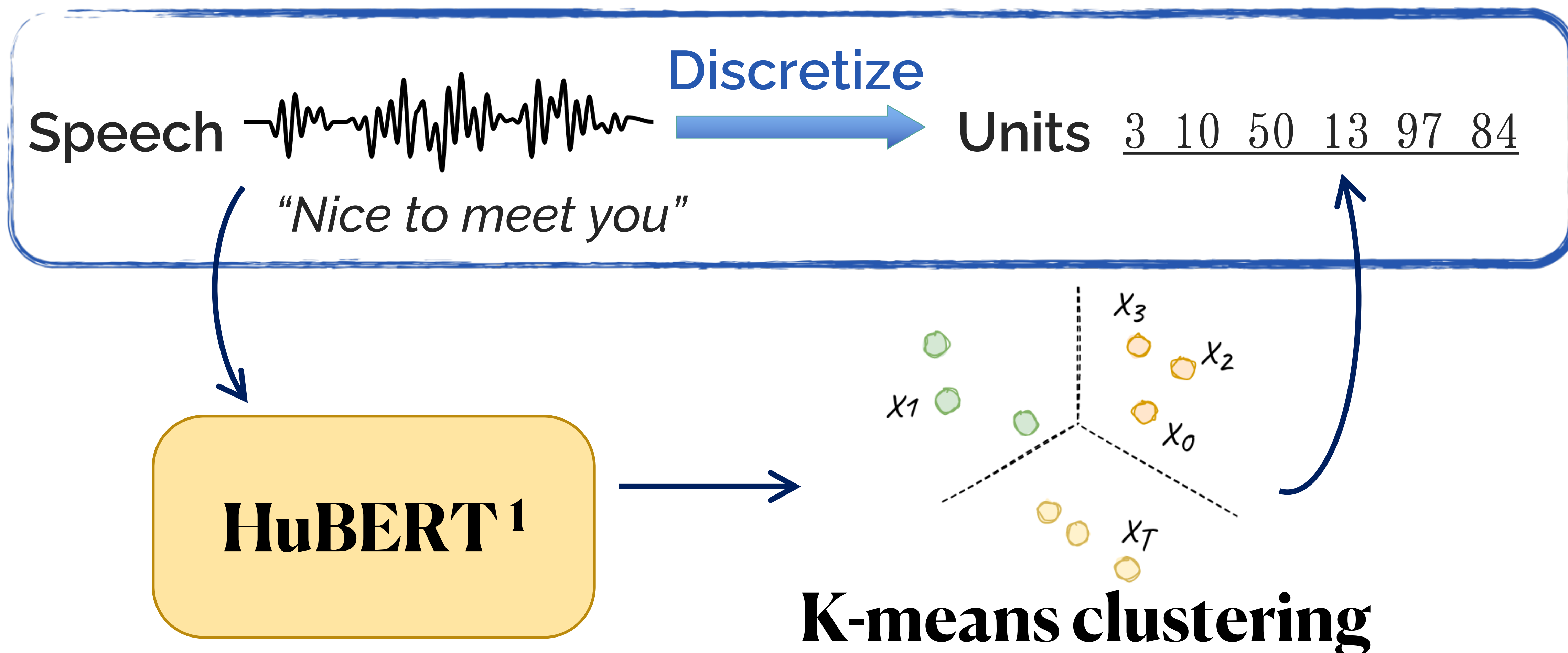
- **Challenge2: Modality Gap**

- hard to **apply low-resource MT** techniques to ST



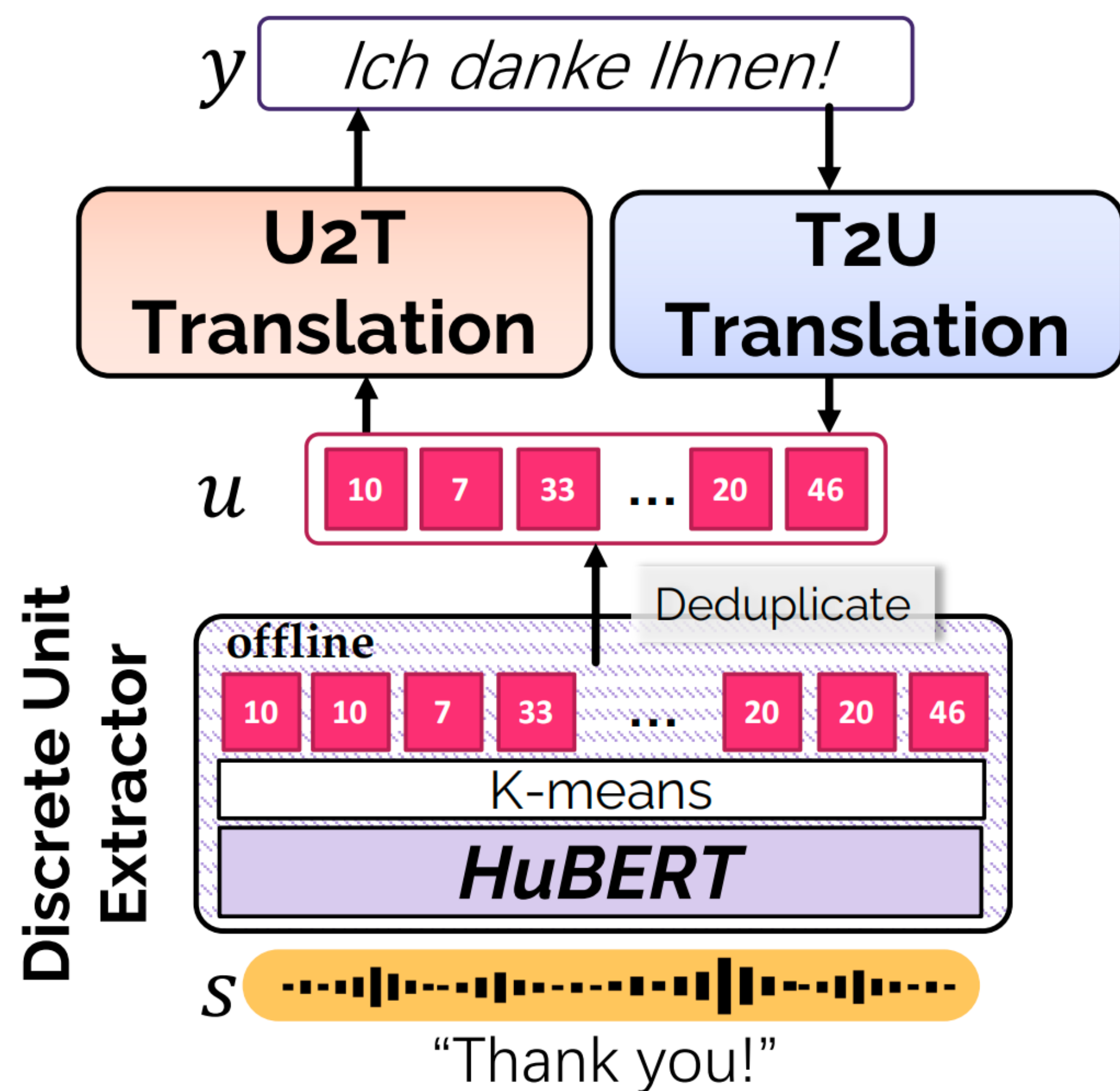


What if Speech can be “Unsupervised” discretized?

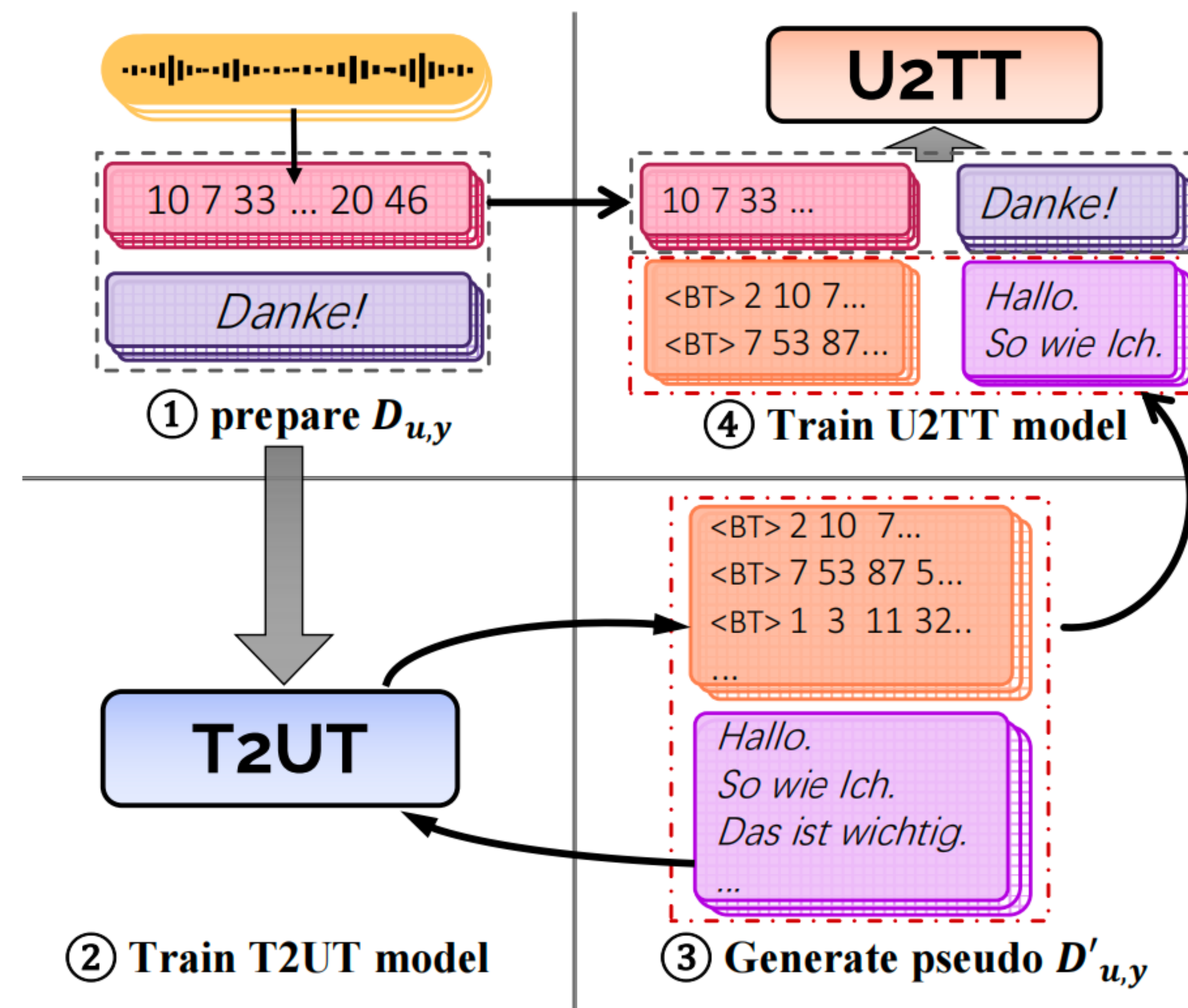




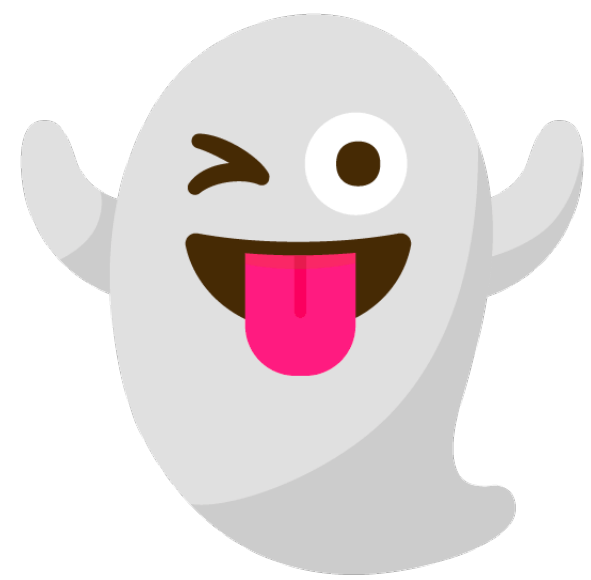
Framework & Training Method



(a) Model Structure



(b) Discrete Unit Back-translation



DUB performs well on both En-X and X-En translation

Method	De	Fr	Es	Avg.
<i>Methods that utilize transcriptions</i>				
Fairseq ST (Wang et al., 2020)	22.7	32.9	27.2	27.6
NeurST (Zhao et al., 2021)	22.8	33.3	27.4	27.8
Espnet ST (Inaguma et al., 2020)	22.9	32.8	28.0	27.9
E2E-ST-JT (Du et al., 2022)	23.1	32.8	27.5	27.8
Speechformer (Papi et al., 2021)	23.6	-	28.5	-
Cascaded (Inaguma et al., 2020)	23.6	33.8	28.7	28.7
MTL (Tang et al., 2021)	23.9	33.1	28.6	28.5
Self-training (Pino et al., 2020)	25.2	34.5	-	-
SpeechT5 (Ao et al., 2022)	25.2	35.3	-	-
<i>Methods that do not involve transcriptions</i>				
Revisit ST (Zhang et al., 2022a)	23.0	33.5	28.0	28.2
Transformer-ST	18.0	28.5	24.1	23.5
U2TT (BASE)	20.4	30.3	25.3	25.3
w/ DUB	25.8	34.7	30.2	30.2
U2TT (LARGE)	20.5	30.1	24.7	25.1
w/ DUB	26.2	35.3	30.4	30.6
<i>SoTA: use much more speech and various pre-training tasks</i>				
SpeechUT (Zhang et al., 2022c)*	30.1	41.4	33.6	35.0

MuST-C
(En-X)

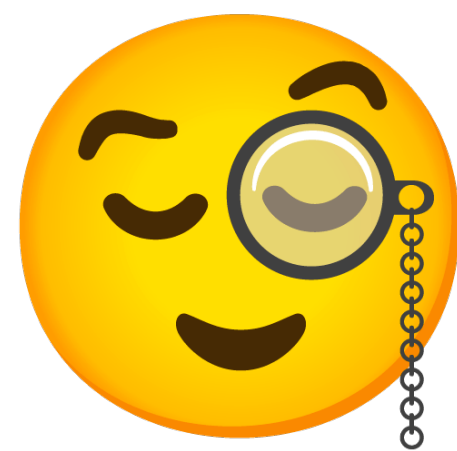
CoVoST-2

(X-En),
low-resource,
unwritten

Methods	Aux. Data ASR	Data Text	Ar 2h	Sv 2h	Lv 2h	Sl 2h	Ta 2h	Ja 1h	Id 1h	Cy 2h	Avg. -
Transformer-ST [†]	-	-	0.3	0.2	0.1	0.3	0.3	0.3	0.4	0.3	0.3
Transformer-ST A2E [†]	-	-	0.6	0.6	0.4	1.2	0.1	0.2	0.3	2.6	0.8
Transformer-ST + ASR pre-train [†]	✓	-	4.3	2.7	2.5	3.0	0.3	1.5	2.5	2.7	2.4
<i>Larger models based on large-scale multilingual speech, text or joint pre-training, involving more data</i>											
XLS-R (0.4B)*	✓	-	8.1	5.3	3.1	5.3	0.0	2.0	3.3	3.4	3.8
Wav2seq (0.4B)*	✓	-	10.5	8.8	4.8	5.9	0.0	1.9	5.0	5.7	5.3
XLS-R + mBART-50 (0.7B) [◇]	✓	✓	3.0	10.3	6.0	6.6	0.2	0.6	1.4	2.5	3.8
LNA-E,D (0.7B) [♣]	✓	✓	3.7	5.9	4.6	4.6	0.7	1.7	2.9	2.8	3.4
U2TT (LARGE)	-	-	7.0	8.0	6.3	6.8	0.3	1.6	6.6	2.7	4.9
w/ DUB	-	✓1M	7.1	8.9	6.9	7.9	0.5	2.1	7.0	5.7	5.8

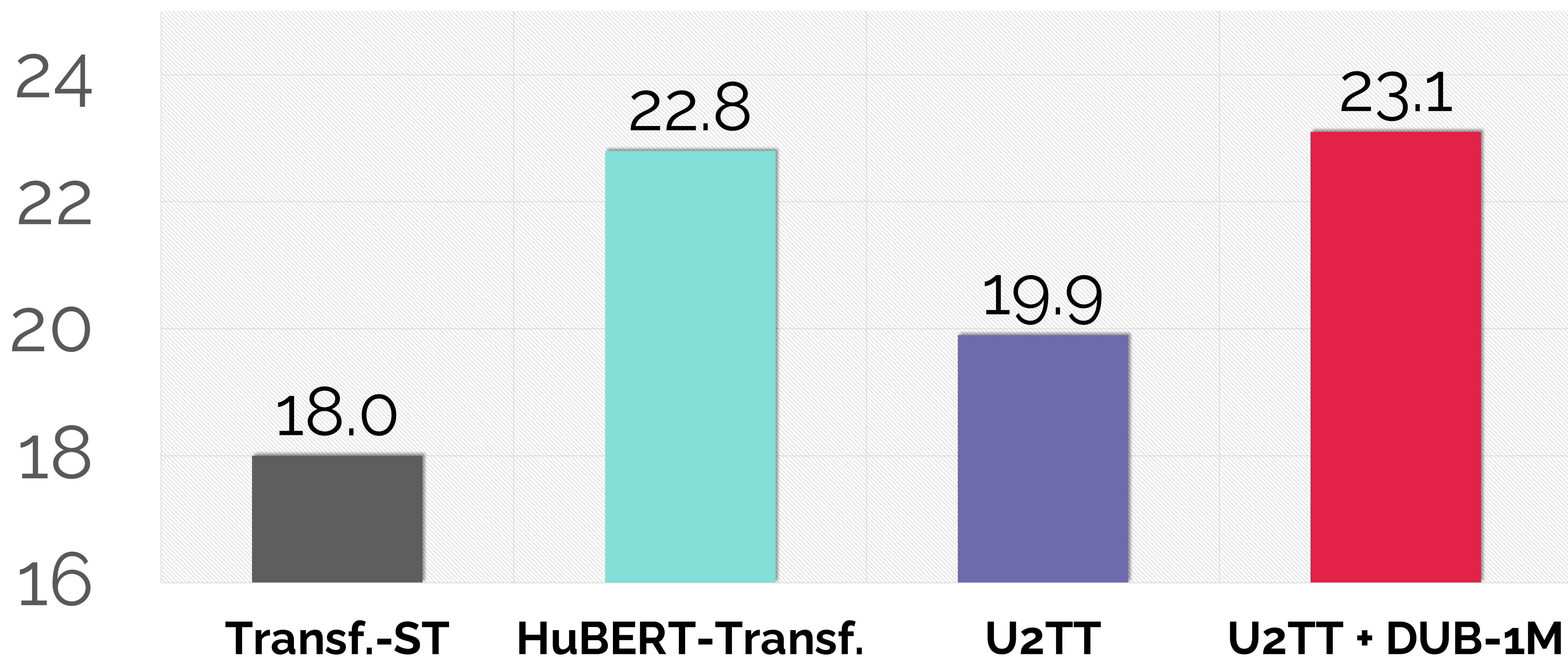
Further investigations: Apply DUB to ST

- 🤔 ① Is **discrete unit** better than **continuous signal** to represent speech for ST?
- 🤔 ② How much benefit can **MT technique**, eg. **BT** bring to direct/E2E ST through discrete unit?



Fbank < Unit < HuBERT continuous

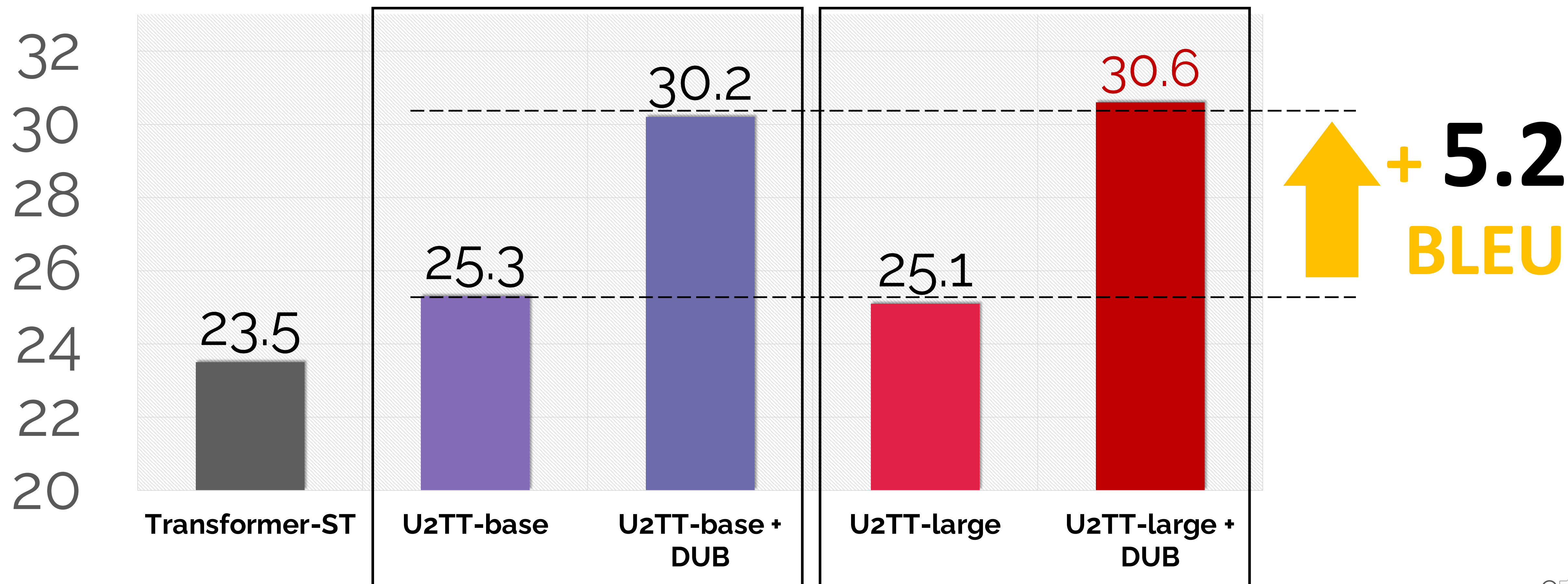
BLEU on MuST-C En-De test

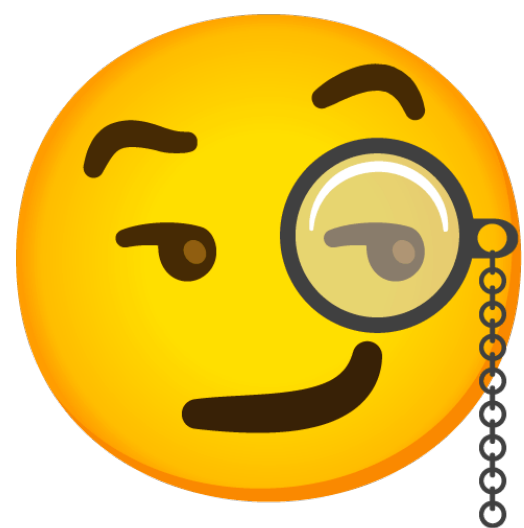




brings **+5 BLEU** for En-X ST

Avg. BLEU on MuST-C En-De/Fr/Es



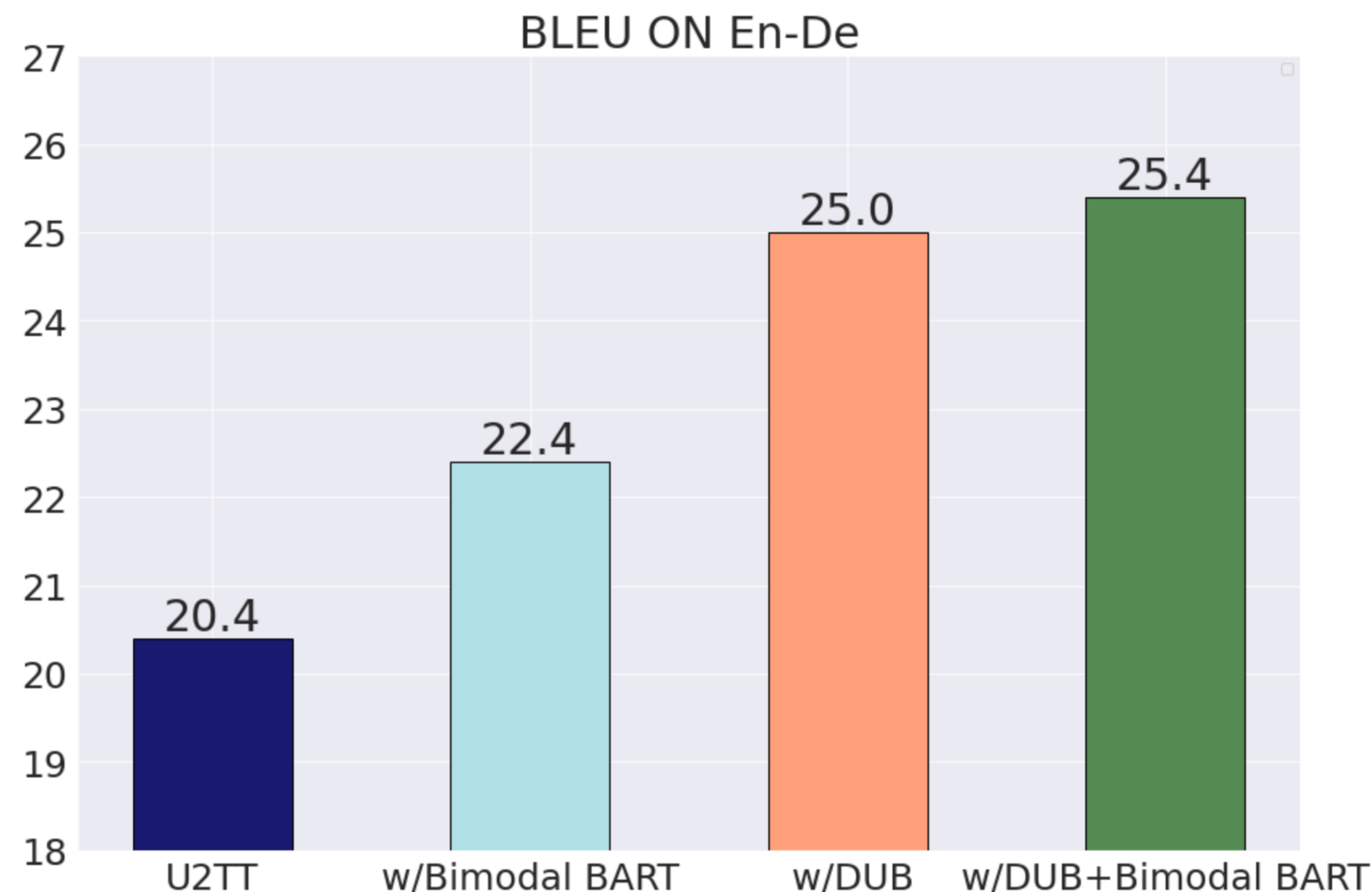


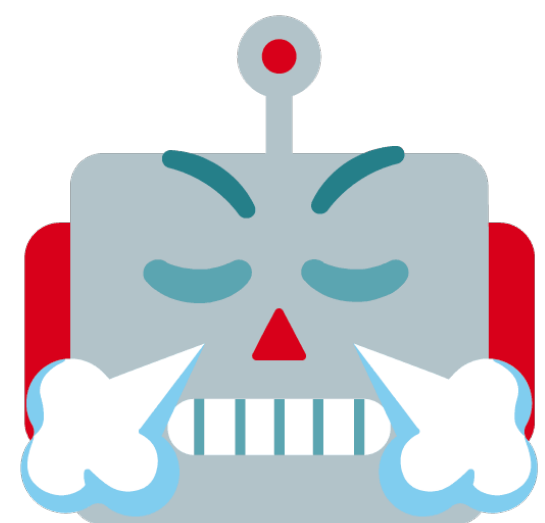
Other techniques: DUB is better than Bimodal BART

- **Bimodal BART**

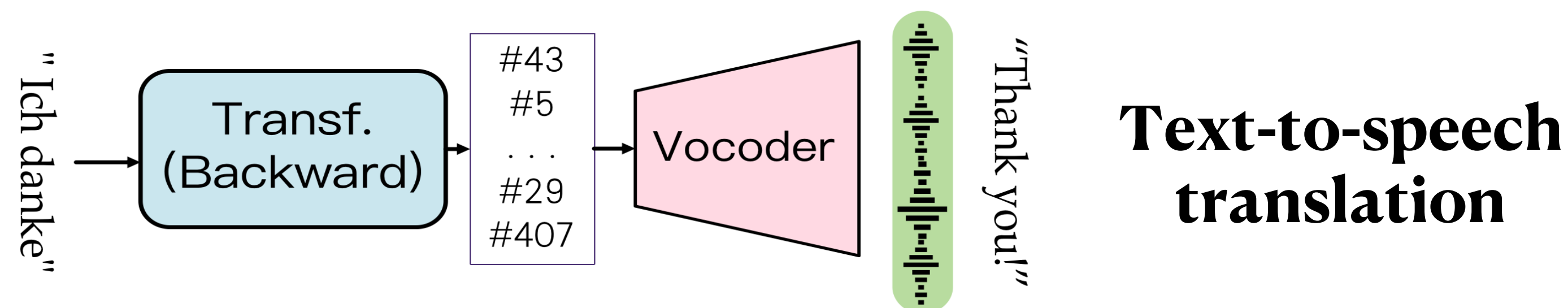
- Denoise large-scale corrupted discrete units and monolingual text like mBART
- Provide a pre-trained model for U2TT

- **Bimodal BART and DUB are complementary**





Speech can be recovered via BT units



German: Und sicher, hätten wir diese diese Hydranten die ganze Zeit freischaufeln können, und viele Menschen tun das.

 **Ref English:** And certainly, we could have been shoveling out these fire hydrants all along, and many people do.

 **BT- Generated English:** And certainly we could have released those hydrants all the time and many people do that.

Take-away of DUB

- Highlight a key advantage of **working with discrete units** for speech and would be of interest to the **broader speech understanding community**.
- Successfully apply **effective MT technique** to direct ST, e.g. **BT**, BART
- Particularly beneficial for **low-resource or unwritten languages**, including translation and preservation.



👉 Thanks



Paper



Code

- **Paper:** <https://arxiv.org/abs/2305.11411>
- **Code:** <https://github.com/onutaton/DUB>
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